Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] [18CIC53]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2021

DATABASE SYSTEMS

Time:3 Hours **Maximum Marks:70**

Instructions: i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.

ii. Missing data may be suitably assumed.

1. a. Explain the characteristics of the Database Approach. **07**

b. Draw ER diagram for Employee data base taking at least 4 entities. **07**

2. a. Write relational algebraic expressions for the following; **06**

i. List all pairs of Employees SSNs and the project numbers they work on

ii. List the employees names along with their department names

iii. List the department names with their managers

b. Demonstrate the uses of the following commands **08**

i. Alter

ii. Insert

iii. Update

iv. Delete

3. a. Explain three tier architecture with its advantages. **07**

b. Explain different types of JDBC Drivers. **07**

OR

4. a. Consider the following schema and write the queries using SQL **10**

EMPLOYEE(Name, SSN, Address, Sex, Salary, Dno)

DEPARTMENT(Dname, Dnumber, MgrSSN, Mgr_Startdate)

i. Retrieve all employees in department 5 whose salary is between 50,000 and 60,000(inclusive)

ii. Retrieve the name and address of all employees who work for the 'Research' department

iii. For each employee, retrieve the employee's name, and the name of his or her immediate supervisor

iv. Increase the salary of all employees working on the 'ProductX' project by 15%

v. Select the names of employees whose salary is more than all the employees of dept 5

b. Explain views in sql with examples. **04**

5.	**a.**	Explain the informal design guidelines for Relational Schemas.	**07**
	b.	What is Normalization? Explain its necessity with example.	**07**
		OR	
6.	**a.**	Explain 3NF in detail.	**07**
	b.	Compare 1 NF and 2NF.	**07**
7.	**a.**	Mention the reasons for transaction fail and highlight the importance of recovery system.	**07**
	b.	Outline the desirable features of a transaction.	**07**
		OR	
8.	**a.**	Give the ARIES recovery algorithm.	**07**
	b.	Compare deadlock prevention and avoidance with examples.	**07**